Question	Answer	Marks	Partial Marks
1(a)(i)	1 | 7 7 8 8 9 9 2 | 1 1 1 1 2 3 3 4 5	2	**B1** for one row correctly ordered or for fully correct unordered stem-and-leaf diagram or for a correct diagram with one error or omission
1(a)(ii)	21	1	
1(a)(iii)	23	1	
1(a)(iv)	48	2	**M1** for $\frac{2}{15}[\times 360]$ or $\frac{360}{15}[\times 2]$
1(b)(i)	120	1	
1(b)(ii)	130	1	
1(b)(iii)	60	1	
1(c)(i)	93.4	4	**M1** for mid-values soi **M1** for Σfx **M1 dep on second M** for $\Sigma fx \div 200$
1(c)(ii)	19	2	**M1** for $\frac{86}{50}$ or $\frac{114}{60}$

Question	Answer	Marks	Partial Marks
2(a)	42	2	**M1** for $12 \div 2$ or better
2(b)(i)	5.72	2	**M1** for $\frac{100-12}{100} \times 6.50$ oe or **B1** for 0.88 oe
2(b)(ii)	12.5[0]	2	**M1** for $\frac{100-12}{100} \times x = 11$ or better oe

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Question	Answer	Marks	Partial Marks
7(a)	39.6 or 39.57....	**4**	**M2** for [cos =] $\frac{14^2+12^2-9^2}{2\times14\times12}$ or **M1** for $9^2 = 14^2 + 12^2 - 2 \times 14 \times 12 \times \cos ACD$ **A1** for 0.7708... or 0.771 or $\frac{37}{48}$ oe
7(b)	$\frac{14\sin 25}{\sin 123}$	**M2**	**M1** for $\frac{\sin 123}{14} = \frac{\sin 25}{BC}$ oe
	7.054…	**A1**	
7(c)	3.74 or 3.735 to 3.739	**3**	**M2** for 7.05 × sin 32 or **M1** for recognition that the line from *B* is perpendicular to *AC*
7(d)	11.8 or 11.83 to 11.85	**4**	**M1** for 32 + *their*(a) soi **M2** for $12^2 + 7.05^2 - 2\times12\times7.05\times\cos(their(a)+32)$ or **M1** for $\cos(their(a)+32) = \frac{12^2+7.05^2-BD^2}{2\times12\times7.05}$
7(e)	309.6 or 309.57...	**2**	**FT** 270 + *their(a)* **M1** for 270 + *their(a)* oe

Question	Answer	Marks	Partial Marks
9(a)	$x^2 - x - 30 [= 0]$	**B3**	**M1** for $(2x+1)(x-1) - x^2 = 29$ oe **B1** for $(2x+1)(x-1) = 2x^2 - 2x + x - 1$ oe soi
	$(x-6)(x+5)$ oe	**M1**	or correct factors for *their* 3 term quadratic equation or for correct substitution into quadratic formula or correctly completing the square for *their* 3 term quadratic equation
	$x = 6$ cao	**B1**	
	12 or 2 × *their x* evaluated or $k = 2x$ stated	**B1 FT**	
9(b)(i)	$(y+1)^3 - y^3 = 5$ oe	**M1**	
	$(y+1)^3 = y^3 + 3y^2 + 3y + 1$ soi	**B2**	**B1** for $(y+1)^2 = y^2 + y + y + 1$ oe soi
	Completion to $3y^2 + 3y - 4 = 0$	**A1**	With no errors or omissions
9(b)(ii)	$\dfrac{-3 \pm \sqrt{3^2 - 4(3)(-4)}}{2 \times 3}$	**B2**	or **B1** for $\sqrt{3^2 - 4(3)(-4)}$ or for $\dfrac{-3 + \sqrt{...}}{2 \times 3}$ or $\dfrac{-3 - \sqrt{...}}{2 \times 3}$
	0.44	**B2**	**B1** for 0.758 or 0.7583...